

Functional requirements

1. The program will read and store data and instructions as words (signed four-digit decimal numbers).
2. The program will store data and instructions in a 100 word memory with addresses from 00 to 99.
3. The program will have an accumulator register.
4. The program will have a current instruction register.
5. The program will load instructions into memory starting at memory address 00.
6. The program will process each instruction sequentially until a HALT (given by the instruction 4300) is encountered.
7. The program may branch depending on the value in the accumulator if a branch instruction (40, 41, or 42) is encountered.
8. The program will display an error message if the input is not an integer.
9. The program can load instructions from files.
10. The program's GUI has a button that allows users to open a file selection window.
11. The program's GUI has a file selection window that allows users which file they would like to load instructions from.
12. The program has a button that allows users to execute the instruction that are loaded in memory.
13. Execution starts back at the beginning each time.
14. The program will wipe its memory once a new set of instructions is executed.
15. The program will wipe the accumulator value once a new set of instructions is executed.

non-functional requirements

1. The program's GUI should be simple and intuitive.
2. The programs should handle errors gracefully.
3. The program will be able to load the instructions and execute them within 5 seconds.